

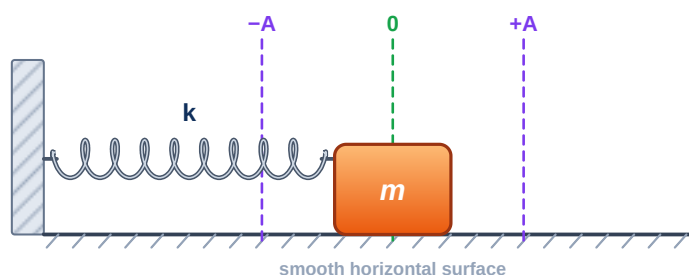
Chapter 2 · Lesson 1

Simple Harmonic Motion

The restoring force · the periodic time and the frequency · the mass–spring oscillator · $x = A \sin \omega t$ ·
energy in SHM
Take $\pi = 3.14$

1

A block of mass 0.5 kg is attached to a horizontal spring of spring constant 50 N/m on a smooth horizontal surface. The block is pulled aside and released, so it oscillates in simple harmonic motion:

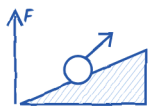


- I The angular frequency of the oscillation equals
- a) 5 rad/s b) 100 rad/s c) 25 rad/s d) 10 rad/s
- II The periodic time of the oscillation equals
- a) 0.628 s b) 0.314 s c) 1.59 s d) 6.28 s

2

The displacement of a body moving in simple harmonic motion is given by the equation $x = 0.05 \sin (20 t)$, where x is in metres and t in seconds:

- I The amplitude of the motion equals
- a) 20 m b) 0.05 m c) 0.1 m d) 1 m
- II The frequency of the motion equals
- a) 20 Hz b) 0.314 Hz c) 3.18 Hz d) 126 Hz



3

Choose the correct answer:

- I** In simple harmonic motion, the restoring force is always
- a) in the direction of motion
 - b) directed towards the equilibrium position and proportional to the displacement
 - c) constant in magnitude
 - d) proportional to the velocity

- II** The speed of the body is maximum at
- a) the equilibrium position
 - b) the two extreme positions
 - c) halfway between them
 - d) every point equally

- III** The acceleration of the body is maximum at
- a) the equilibrium position
 - b) halfway to the extremes
 - c) the two extreme positions
 - d) no point

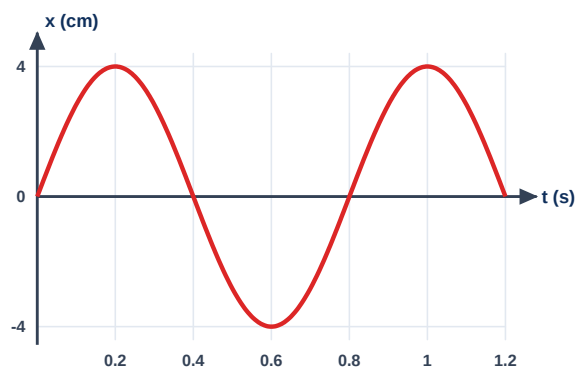
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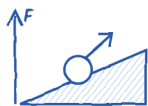
The graph shows how the displacement of a body in simple harmonic motion changes with time:

- I** The amplitude of the motion is
- a) 8 cm
 - b) 4 cm
 - c) 2 cm
 - d) 0.8 cm

- II** The frequency of the motion is
- a) 0.8 Hz
 - b) 2.5 Hz
 - c) 1.25 Hz
 - d) 7.85 Hz

- III** Calculate the angular frequency.



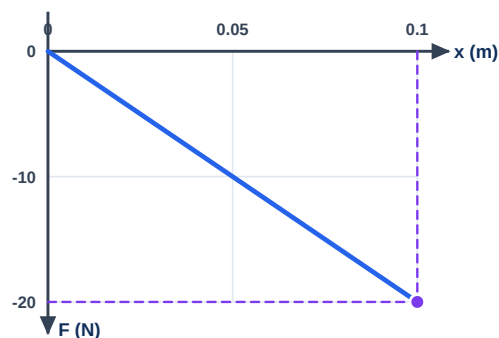


5

The graph shows the restoring force F acting on a body attached to a spring against its displacement x from the equilibrium position:

I The spring constant equals

- a) 2 N/m b) 20 N/m
c) 200 N/m d) 2000 N/m



II If the mass of the body is 2 kg, its periodic time is

- a) 0.628 s b) 0.314 s
c) 6.28 s d) 0.1 s

III What does the negative slope of the line mean?

6

A body oscillates in simple harmonic motion with an amplitude of 8 cm and a periodic time of 0.4 s:

I Its maximum speed equals

- a) 1.256 m/s b) 0.2 m/s c) 3.14 m/s d) 19.7 m/s

II Calculate its maximum acceleration.

III The ratio $a_{\max} / v_{\max}$ equals

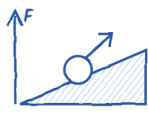
- a) T b) A c) ω d) f

7

A body in simple harmonic motion has a maximum speed of 2 m/s and a maximum acceleration of 40 m/s^2 :

I Its angular frequency equals

- a) 80 rad/s b) 40 rad/s c) 0.05 rad/s d) 20 rad/s



I Find the amplitude and the periodic time.

8

A body of mass m attached to a horizontal spring oscillates with a periodic time of 0.6 s :

I If another mass of $3m$ is fixed to the body, the new periodic time is

- a) 0.3 s b) 0.6 s c) 2.4 s d) 1.2 s

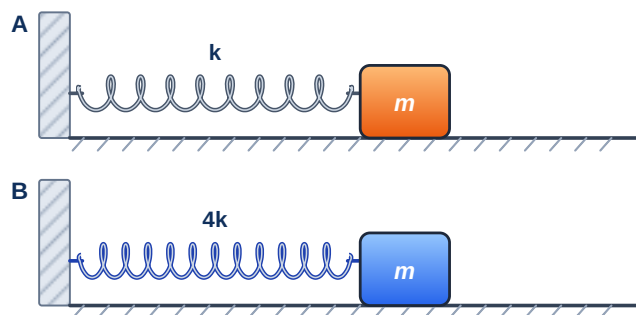
II To make the periodic time 0.3 s with the same spring, the mass must be

- a) $m / 2$ b) $m / 4$ c) $2m$ d) $4m$

III With the original mass, what is the periodic time if the spring is replaced by one 4 times stiffer?

9

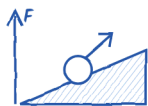
Two identical bodies, each of mass m , oscillate horizontally on two springs: spring A of constant k and spring B of constant $4k$, as shown:



I The ratio of the periodic times T_A / T_B equals

- a) 4 b) $1/2$ c) $1/4$ d) 2

II What mass must be fixed to spring B so that its periodic time equals that of A?



10

A body of mass 0.2 kg oscillates on a horizontal spring of constant 80 N/m with an amplitude of 10 cm :

I The total mechanical energy of the oscillator is

a) 0.8 J

b) 0.04 J

c) 4 J

d) 0.4 J

II Calculate the maximum speed of the body.

III Calculate its speed when the displacement is 5 cm .

11

The graph shows two energy curves (1) and (2) of a body in simple harmonic motion against its displacement x , together with the total energy E :

I The curve that represents the kinetic energy is

a) curve (1)

b) curve (2)

c) the line E

d) none of them

II At point P the kinetic and potential energies are equal. Its displacement is

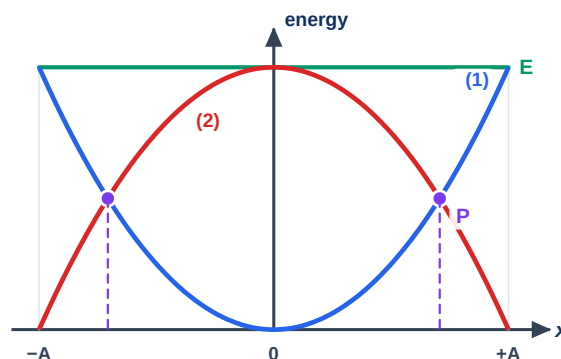
a) $A/2$

b) $A/4$

c) $A/\sqrt{2}$

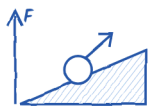
d) A

III Find the ratio $KE : PE$ when $x = A/2$.



5

Block 1



12

A body oscillates in simple harmonic motion with a periodic time of 1.2 s, starting from the equilibrium position ($x = A \sin \omega t$):

I The shortest time taken to move from $x = 0$ to $x = A/2$ is

- a) 0.3 s b) 0.15 s c) 0.1 s d) 0.2 s

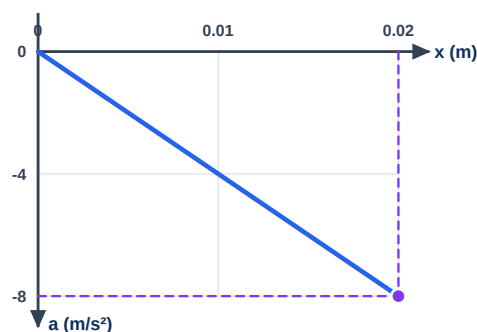
II Find the time taken to move from $x = A/2$ to $x = A$, and explain why it is longer than the first half.

13

The graph shows the acceleration a of a body in simple harmonic motion against its displacement x :

I The angular frequency of the motion is

- a) 400 rad/s b) 4 rad/s
c) 0.05 rad/s d) 20 rad/s



II Calculate the periodic time.

III What property of simple harmonic motion does this graph show?

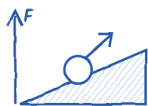
14

The displacement of a body in simple harmonic motion is $x = 0.1 \sin (5\pi t)$, where x is in metres and t in seconds:

I The displacement of the body at $t = 0.1$ s is

- a) zero b) 0.05 m c) 0.1 m d) -0.1 m

6

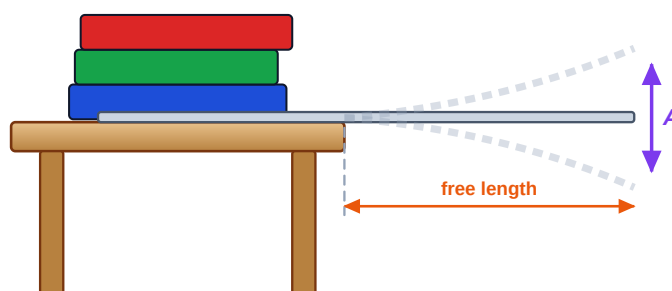


I Find the velocity and the acceleration of the body at that instant.

III Find the periodic time.

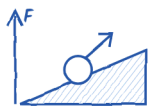
15

A steel ruler is clamped at the edge of a table and its free end is set vibrating, as shown:



I As the amplitude of the vibration slowly decreases, the periodic time
a) increases b) decreases c) remains constant d) becomes zero

I If the free length of the ruler is made shorter, what happens to the frequency? Explain.



18

The spring of a mass-spring oscillator is replaced by one 4 times stiffer, and the mass is doubled:

I The new periodic time T' in terms of the old one T is

- a) $T / \sqrt{2}$ b) $T / 2$ c) $T \sqrt{2}$ d) $2T$

II How does the frequency change?

19

A body of mass 0.2 kg is acted upon by a force F that depends on its displacement x :

I The body moves in simple harmonic motion if the force is

- a) $F = -5x^2$ b) $F = 5x$ c) $F = -5x$ d) $F = -5$

II For this force, find the angular frequency and the periodic time.

20

The graph shows the displacement of a body in simple harmonic motion against time. P, Q, R, S and T are five points on the curve:

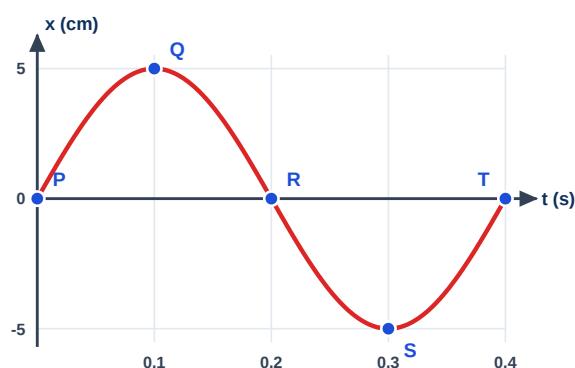
I The speed of the body is maximum at

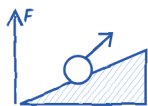
- a) P, R and T b) Q and S
c) Q only d) S only

II The acceleration is maximum and positive at

- a) Q b) S
c) R d) P

III At which points is the restoring force zero?





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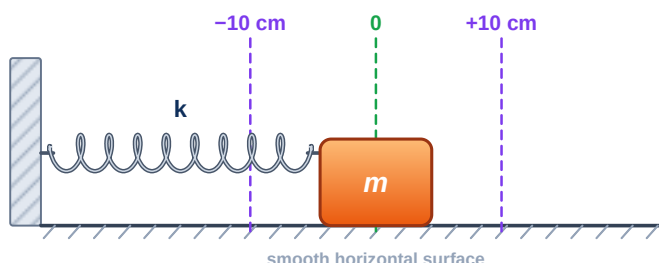
A body oscillates in simple harmonic motion with an angular frequency of 10 rad/s :

- I When its displacement is -3 cm , its acceleration is
- a) -3 m/s^2 b) $+30 \text{ m/s}^2$ c) -0.3 m/s^2 d) $+3 \text{ m/s}^2$

- II If the mass of the body is 0.5 kg , find the restoring force when $x = +3 \text{ cm}$.

22

A body of mass 2 kg attached to a horizontal spring of constant 200 N/m is pulled 10 cm from its equilibrium position and released:



- I The restoring force at the moment of release is
- a) 2 N b) 20 N c) 200 N d) 2000 N

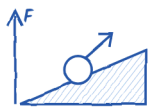
- II Find its acceleration at the moment of release.

- III Find its speed as it passes the equilibrium position.

23

A body oscillates with a frequency of 2.5 Hz and an amplitude of 5 cm , starting from the equilibrium position:

- I The number of complete oscillations it makes in one minute is
- a) 25 b) 150 c) 2.5 d) 60



I Find the distance it travels in 2 s, and its displacement at the end of these 2 s.

24

A body oscillates with an amplitude of 10 cm and an angular frequency of 10 rad/s:

I Its speed when its displacement is 6 cm equals

a) 1 m/s

b) 0.6 m/s

c) 0.8 m/s

d) 0.36 m/s

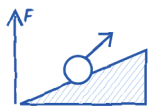
I At what displacement is its speed half the maximum speed?

25

The speed of a body in simple harmonic motion is 0.8 m/s when its displacement is 6 cm, and 0.6 m/s when its displacement is 8 cm:

I Find the angular frequency of the motion.

I Find the amplitude.



26

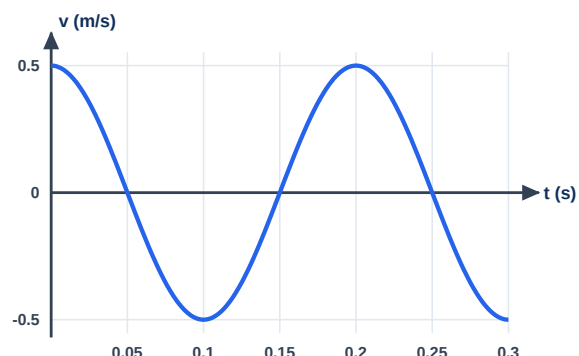
The graph shows how the velocity of a body in simple harmonic motion changes with time:

I Find the angular frequency.

II The amplitude of the motion is

- a) 1.59 cm b) 15.7 cm
c) 0.1 cm d) 3.14 cm

III Where is the body at $t = 0$?



27

Choose the correct answer:

I When the displacement of a body in simple harmonic motion is $A/3$, the ratio $KE : PE$ equals

- a) 3 b) $1/8$ c) 9 d) 8

II A body of mass 0.5 kg oscillates with an amplitude of 4 cm and a frequency of 5 Hz. The maximum restoring force on it is

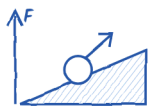
- a) 19.7 N b) 9.86 N c) 39.4 N d) 0.63 N

28

A student finds that a body of unknown mass m_1 on a spring oscillates with a periodic time of 0.5 s. When a mass of 0.3 kg is added, the periodic time becomes 0.7 s:

I Find the mass m_1 .

II Find the spring constant.

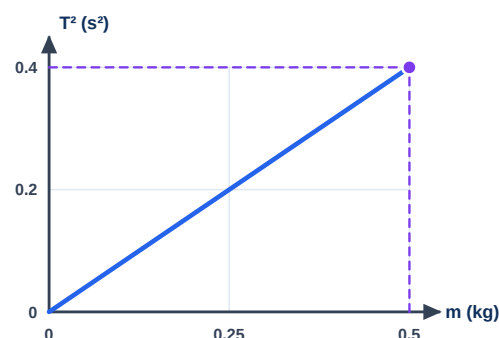


29

The graph shows the square of the periodic time T^2 of a mass-spring oscillator against the mass m :

I The spring constant equals

- a) 49.3 N/m b) 0.8 N/m
c) 24.6 N/m d) 98.6 N/m



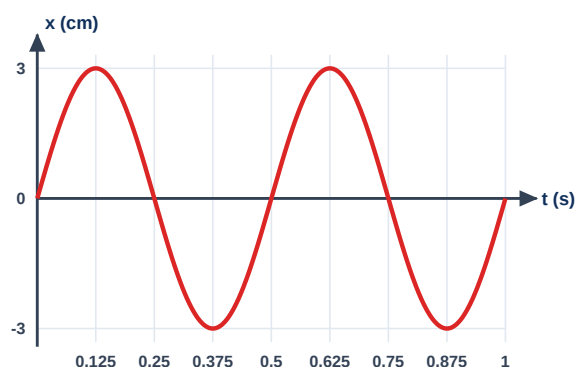
II Find the periodic time when a mass of 0.8 kg is used.

30

The graph shows the displacement-time curve of a body in simple harmonic motion:

I The angular frequency of the motion is

- a) 6.28 rad/s b) 3.14 rad/s
c) 12.56 rad/s d) 2 rad/s



II Write the equation of the displacement in metres.

31

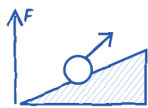
A body starts its simple harmonic motion from the equilibrium position ($x = A \sin \omega t$):

I At $t = T/8$ the ratio KE : PE equals

- a) 1 : 1 b) 1 : 3 c) 3 : 1 d) 1 : 2

II A body oscillates with an amplitude of 2.5 cm and a maximum acceleration of 10 m/s^2 . Its frequency is

- a) 3.18 Hz b) 20 Hz c) 0.314 Hz d) 400 Hz



32

Give the scientific reason for each of the following:

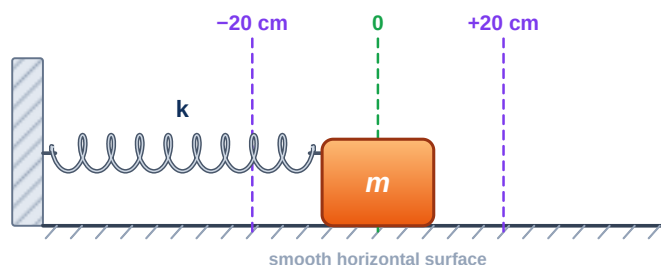
I The periodic time of a mass-spring oscillator does not depend on the amplitude.

II The speed of the oscillating body is maximum at the equilibrium position.

III A horizontal mass-spring oscillator has the same periodic time on the Moon as on the Earth.

33

A body of mass 1 kg oscillates on a horizontal spring of constant 100 N/m with an amplitude of 20 cm:



I Its speed at a displacement of 12 cm equals

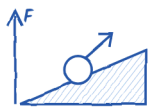
a) 2 m/s

b) 1.6 m/s

c) 1.2 m/s

d) 0.8 m/s

II Find the kinetic and potential energies at that point, and check that their sum is constant.



34

The total energy of a mass-spring oscillator is 0.9 J. The mass is 0.2 kg and the amplitude is 15 cm:

I Find the maximum speed.

II Find the spring constant.

III The periodic time equals

- a) 0.314 s b) 0.628 s c) 0.05 s d) 3.14 s

35

Choose the correct answer:

I If the mass of a mass-spring oscillator is made 4 times as large, its frequency becomes

- a) $4f$ b) $2f$ c) $f/2$ d) $f/4$

II The SI unit of the spring constant is

- a) N·m b) N/m c) N/s d) N

III During one complete oscillation, the displacement of the body is

- a) $4A$ b) $2A$ c) A d) zero

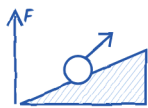
36

The displacement of a body is $x = 6 \sin(\pi t)$, where x is in cm and t in seconds:

I The first moment at which $x = 3$ cm is

- a) 0.5 s b) $1/6$ s c) $1/3$ s d) $1/12$ s

II Find the speed of the body at that moment.



37

A body in simple harmonic motion passes its equilibrium position with a speed of 0.6 m/s . Its amplitude is 3 cm :

- I Its angular frequency equals
- a) 0.2 rad/s b) 2 rad/s c) 18 rad/s d) 20 rad/s

II Find its periodic time.

III Find its acceleration when its displacement is 1.5 cm .

38

A body of mass 0.4 kg is attached to a horizontal spring, pulled 5 cm from its equilibrium position and released. It makes 20 complete oscillations in 8 s :

- I The periodic time equals
- a) 2.5 s b) 0.8 s c) 160 s d) 0.4 s

II Find the spring constant.

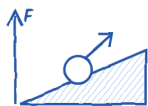
III Find its maximum speed.

39

A body in simple harmonic motion has an acceleration of magnitude 8 m/s^2 when its displacement is 2 cm :

- I Its angular frequency equals
- a) 400 rad/s b) 4 rad/s c) 2 rad/s d) 20 rad/s

II At what displacement is the magnitude of its acceleration 12 m/s^2 ?



III Find the periodic time.

40

Two bodies, each of mass 1 kg, oscillate on two springs of constants $k_1 = 100 \text{ N/m}$ and $k_2 = 400 \text{ N/m}$, both with the same amplitude of 5 cm:

I The ratio of their frequencies f_1 / f_2 equals

a) 2

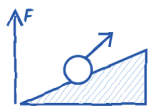
b) 4

c) 1/4

d) 1/2

II Find the ratio of their total energies E_1 / E_2 .

III Find the ratio of their maximum speeds.



Chapter 2 · Lesson 1

Model Answers

Every question solved step by step

1

$$\omega = \sqrt{k / m} = \sqrt{(50 / 0.5)} = \sqrt{100}$$

$$\omega = 10 \text{ rad/s}$$

d) 10 rad/s

$$\text{II } T = 2\pi / \omega = (2 \times 3.14) / 10$$

$$T = 0.628 \text{ s}$$

a) 0.628 s

2

Comparing with $x = A \sin(\omega t)$:

$$A = 0.05 \text{ m}$$

b) 0.05 m

$$\text{II } \omega = 20 \text{ rad/s}$$

$$f = \omega / 2\pi = 20 / 6.28 = 3.18 \text{ Hz}$$

c) 3.18 Hz

3

$F = -kx$: proportional to the displacement x ,
and the negative sign means it points towards $x = 0$.

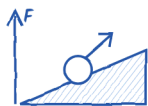
b) directed towards the equilibrium position and proportional to the displacement

II At $x = 0$ all the energy is kinetic, so the speed is maximum : $v_{\max} = \omega A$.

a) the equilibrium position

III $a = -\omega^2 x$: its magnitude is largest where $|x|$ is largest, at $x = \pm A$.

c) the two extreme positions



4

The largest displacement read from the graph is 4 cm.

$A = 4 \text{ cm}$

b) 4 cm

I One complete cycle takes $T = 0.8 \text{ s}$

$f = 1 / T = 1 / 0.8 = 1.25 \text{ Hz}$

c) 1.25 Hz

III $\omega = 2\pi / T = 6.28 / 0.8 = 7.85 \text{ rad/s}$

$\omega = 7.85 \text{ rad/s}$

5

$F = -kx \rightarrow k = |F| / x$

$k = 20 / 0.1 = 200 \text{ N/m}$

c) 200 N/m

I $T = 2\pi \sqrt{m / k} = 6.28 \times \sqrt{2 / 200}$

$T = 6.28 \times 0.1 = 0.628 \text{ s}$

a) 0.628 s

III The restoring force is always opposite to the displacement ($F = -kx$):
it always pulls the body back towards the equilibrium position.

F is always opposite to x

6

$\omega = 2\pi / T = 6.28 / 0.4 = 15.7 \text{ rad/s}$

$v_{\max} = \omega A = 15.7 \times 0.08 = 1.256 \text{ m/s}$

a) 1.256 m/s

I $a_{\max} = \omega^2 A = (15.7)^2 \times 0.08$

$a_{\max} = 246.5 \times 0.08 = 19.7 \text{ m/s}^2$

$a_{\max} \approx 19.7 \text{ m/s}^2$

III $a_{\max} / v_{\max} = \omega^2 A / \omega A = \omega = 15.7 \text{ rad/s}$

c) ω

7

$a_{\max} / v_{\max} = \omega^2 A / \omega A = \omega$

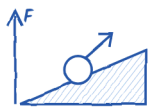
$\omega = 40 / 2 = 20 \text{ rad/s}$

d) 20 rad/s

I $A = v_{\max} / \omega = 2 / 20 = 0.1 \text{ m}$

$T = 2\pi / \omega = 6.28 / 20 = 0.314 \text{ s}$

$A = 0.1 \text{ m}, T = 0.314 \text{ s}$



8

$$T = 2\pi \sqrt{m/k} \rightarrow T \propto \sqrt{m}$$

the mass becomes $4m \rightarrow T' = \sqrt{4} \times 0.6 = 1.2 \text{ s}$

d) 1.2 s

I $T' / T = 0.3 / 0.6 = 1/2 = \sqrt{m' / m}$

$$m' / m = 1/4 \rightarrow m' = m / 4$$

b) $m / 4$

III $T \propto 1 / \sqrt{k} \rightarrow T' = T / \sqrt{4} = 0.6 / 2 = 0.3 \text{ s}$

$T' = 0.3 \text{ s}$

9

$$T \propto 1 / \sqrt{k} \rightarrow T_A / T_B = \sqrt{4k / k} = 2$$

d) 2

I $T_B = 2\pi \sqrt{M / 4k}$ must equal $T_A = 2\pi \sqrt{m / k}$

$$M / 4k = m / k \rightarrow M = 4m$$

$M = 4m$

10

$$E = \frac{1}{2} k A^2 = \frac{1}{2} \times 80 \times (0.1)^2$$

$$E = 0.4 \text{ J}$$

d) 0.4 J

I At $x = 0$ all the energy is kinetic : $\frac{1}{2} m v_{\max}^2 = E$

$$v_{\max} = \sqrt{2E / m} = \sqrt{2 \times 0.4 / 0.2} = 2 \text{ m/s}$$

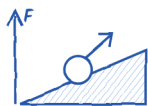
$v_{\max} = 2 \text{ m/s}$

III $PE = \frac{1}{2} k x^2 = \frac{1}{2} \times 80 \times (0.05)^2 = 0.1 \text{ J}$

$$KE = E - PE = 0.4 - 0.1 = 0.3 \text{ J}$$

$$v = \sqrt{2 KE / m} = \sqrt{2 \times 0.3 / 0.2} = \sqrt{3} \approx 1.73 \text{ m/s}$$

$v \approx 1.73 \text{ m/s}$



11

The kinetic energy is maximum at $x = 0$ and zero at $x = \pm A \rightarrow$ curve (2).
Curve (1) is the potential energy $\frac{1}{2}kx^2$, zero at $x = 0$.

b) curve (2)

$$\text{I} \quad \text{PE} = \text{KE} = E/2 \rightarrow \frac{1}{2} k x^2 = \frac{1}{2} (\frac{1}{2} k A^2) \\ x^2 = A^2 / 2 \rightarrow x = A / \sqrt{2} \approx 0.71 A$$

c) $A / \sqrt{2}$

$$\text{III} \quad \text{PE} = \frac{1}{2} k (A/2)^2 = E / 4 \\ \text{KE} = E - E/4 = 3E / 4 \\ \text{KE} : \text{PE} = 3 : 1$$

KE : PE = 3 : 1

12

$$A/2 = A \sin(\omega t) \rightarrow \sin(\omega t) = 0.5 \rightarrow \omega t = \pi/6 \\ t = (\pi/6) / (2\pi/T) = T / 12 = 1.2 / 12 = 0.1 \text{ s}$$

c) 0.1 s

$$\text{I} \quad \text{from } 0 \text{ to } A \text{ takes } T/4 = 0.3 \text{ s} \\ \text{from } A/2 \text{ to } A : 0.3 - 0.1 = 0.2 \text{ s} \\ \text{the body slows down as it approaches the extreme position, so the second half takes longer}$$

t = 0.2 s

13

$$a = -\omega^2 x \rightarrow \omega^2 = |a| / x = 8 / 0.02 = 400 \\ \omega = 20 \text{ rad/s}$$

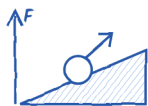
d) 20 rad/s

$$\text{I} \quad T = 2\pi / \omega = 6.28 / 20 = 0.314 \text{ s}$$

T = 0.314 s

$$\text{III} \quad \text{The acceleration is proportional to the displacement and always opposite to it } (a = -\omega^2 x).$$

$a \propto -x$



14

$$x = 0.1 \sin(5\pi \times 0.1) = 0.1 \sin(\pi/2)$$

$$x = 0.1 \times 1 = 0.1 \text{ m (an extreme position)}$$

c) 0.1 m

I At an extreme position the body stops for an instant : $v = 0$

$$a = -\omega^2 x = -(5 \times 3.14)^2 \times 0.1 = -246.5 \times 0.1 \approx -24.6 \text{ m/s}^2$$

$$v = 0, a \approx -24.6 \text{ m/s}^2$$

III $\omega = 5\pi \rightarrow T = 2\pi / \omega = 2\pi / 5\pi = 0.4 \text{ s}$

$$T = 0.4 \text{ s}$$

15

The periodic time of simple harmonic motion does not depend on the amplitude:
that is why the pitch of the sound stays the same.

c) remains constant

I A shorter free length is stiffer (a larger k),

$$f = (1 / 2\pi) \sqrt{(k / m)} \rightarrow \text{the frequency increases (a higher pitch).}$$

the frequency increases

16

Both take $T/4$, and $T = 2\pi \sqrt{(m / k)}$ does not depend on the amplitude.

c) both at the same time

I $v_{\max} = \omega A$, with the same ω

$$v_A : v_B = 2 : 5$$

$$2 : 5$$

III $E = \frac{1}{2} k A^2 \rightarrow E \propto A^2$

$$E_A : E_B = 2^2 : 5^2 = 4 : 25$$

$$4 : 25$$

17

$T = 2\pi \sqrt{(m/k)}$: unchanged

$v_{\max} = \omega A$: doubled

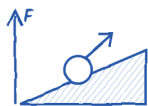
$$E = \frac{1}{2} k A^2 : \times 4$$

$a_{\max} = \omega^2 A$: doubled

b) the maximum speed is doubled

I $E \propto A^2 \rightarrow E' / E = 2^2 = 4$

c) 4 times



18

$$T' / T = \sqrt{(m' / m)} \times \sqrt{(k / k')} = \sqrt{2} \times \sqrt{(1/4)}$$

$$T' / T = \sqrt{2} / 2 = 1 / \sqrt{2}$$

a) $T / \sqrt{2}$

II $f \propto 1 / T \rightarrow f' = \sqrt{2} f \approx 1.41 f$

$f' = \sqrt{2} f$

19

SHM needs a force proportional to x and opposite to it : $F = -k x$.

c) $F = -5 x$

II $k = 5 \text{ N/m}$

$$\omega = \sqrt{(k / m)} = \sqrt{(5 / 0.2)} = \sqrt{25} = 5 \text{ rad/s}$$

$$T = 2\pi / \omega = 6.28 / 5 = 1.256 \text{ s}$$

$\omega = 5 \text{ rad/s}$, $T = 1.256 \text{ s}$

20

The speed is maximum where $x = 0$ (the steepest parts of the curve) : P, R and T.

a) P, R and T

II $a = -\omega^2 x$ is largest and positive where $x = -A$: point S.

b) S

III $F = -k x = 0$ where $x = 0$: P, R and T.

P, R and T

21

$$a = -\omega^2 x = -(10)^2 \times (-0.03)$$

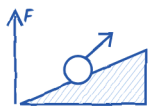
$$a = +3 \text{ m/s}^2 \text{ (towards the equilibrium position)}$$

d) $+3 \text{ m/s}^2$

II $a = -\omega^2 x = -100 \times 0.03 = -3 \text{ m/s}^2$

$$F = m a = 0.5 \times (-3) = -1.5 \text{ N}$$

$F = -1.5 \text{ N}$



22

$$|F| = kx = 200 \times 0.1 = 20 \text{ N (towards the equilibrium position)}$$

b) 20 N

I $a = F / m = 20 / 2 = 10 \text{ m/s}^2$

$a = 10 \text{ m/s}^2$

III $\omega = \sqrt{(k / m)} = \sqrt{(200 / 2)} = 10 \text{ rad/s}$

$$v_{\max} = \omega A = 10 \times 0.1 = 1 \text{ m/s}$$

$v = 1 \text{ m/s}$

23

$$N = f \times t = 2.5 \times 60 = 150 \text{ oscillations}$$

b) 150

I in 2 s it makes $2.5 \times 2 = 5$ complete oscillations

each oscillation covers $4A = 20 \text{ cm}$

distance = $5 \times 20 = 100 \text{ cm} = 1 \text{ m}$

after whole oscillations it is back where it started : displacement = 0

distance = 1 m , displacement = 0

24

energy : $\frac{1}{2} k A^2 = \frac{1}{2} k x^2 + \frac{1}{2} m v^2 \rightarrow v = \omega \sqrt{(A^2 - x^2)}$

$$v = 10 \times \sqrt{(0.01 - 0.0036)} = 10 \times 0.08 = 0.8 \text{ m/s}$$

c) 0.8 m/s

I $v = v_{\max} / 2 \rightarrow \omega \sqrt{(A^2 - x^2)} = \omega A / 2$

$$A^2 - x^2 = A^2 / 4 \rightarrow x = (\sqrt{3} / 2) A$$

$$x = 0.866 \times 10 = 8.66 \text{ cm}$$

$x \approx 8.66 \text{ cm}$

25

$v^2 = \omega^2 (A^2 - x^2)$ for both positions

$$0.64 = \omega^2 (A^2 - 0.0036) \text{ and } 0.36 = \omega^2 (A^2 - 0.0064)$$

subtracting : $0.28 = \omega^2 \times 0.0028 \rightarrow \omega^2 = 100 \rightarrow \omega = 10 \text{ rad/s}$

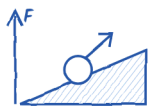
$\omega = 10 \text{ rad/s}$

I $A^2 = 0.64 / 100 + 0.0036 = 0.0064 + 0.0036 = 0.01$

$$A = 0.1 \text{ m} = 10 \text{ cm}$$

$A = 10 \text{ cm}$

24



26

$T = 0.2 \text{ s}$ (one complete cycle)

$$\omega = 2\pi / T = 6.28 / 0.2 = 31.4 \text{ rad/s}$$

$$\omega = 31.4 \text{ rad/s}$$

I $v_{\text{max}} = \omega A \rightarrow A = 0.5 / 31.4 = 0.0159 \text{ m} \approx 1.59 \text{ cm}$

$$\text{a) } 1.59 \text{ cm}$$

III At $t = 0$ the speed is maximum, so the body is at the equilibrium position.

at the equilibrium position

27

$$PE = \frac{1}{2} k (A/3)^2 = E / 9$$

$$KE = E - E/9 = 8E / 9$$

$$KE : PE = 8$$

$$\text{d) } 8$$

I $\omega = 2\pi f = 31.4 \text{ rad/s}$

$$F_{\text{max}} = m \omega^2 A = 0.5 \times 986 \times 0.04 \approx 19.7 \text{ N}$$

$$\text{a) } 19.7 \text{ N}$$

28

$$T^2 \propto m \rightarrow (0.7 / 0.5)^2 = (m_1 + 0.3) / m_1$$

$$1.96 m_1 = m_1 + 0.3$$

$$0.96 m_1 = 0.3 \rightarrow m_1 \approx 0.31 \text{ kg}$$

$$m_1 \approx 0.31 \text{ kg}$$

I $k = 4\pi^2 m_1 / T^2 = 39.44 \times 0.3125 / 0.25$

$$k \approx 49.3 \text{ N/m}$$

$$k \approx 49.3 \text{ N/m}$$

29

$$T^2 = (4\pi^2 / k) m \rightarrow \text{slope} = 4\pi^2 / k$$

$$\text{slope} = 0.4 / 0.5 = 0.8 \text{ s}^2/\text{kg}$$

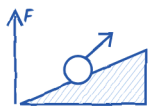
$$k = 4\pi^2 / 0.8 = 39.44 / 0.8 \approx 49.3 \text{ N/m}$$

$$\text{a) } 49.3 \text{ N/m}$$

I $T^2 = 0.8 \times 0.8 = 0.64 \text{ s}^2$

$$T = 0.8 \text{ s}$$

$$T = 0.8 \text{ s}$$



30

$$T = 0.5 \text{ s}$$

$$\omega = 2\pi / T = 6.28 / 0.5 = 12.56 \text{ rad/s}$$

c) 12.56 rad/s

II $A = 3 \text{ cm} = 0.03 \text{ m}$, $\omega = 12.56 \text{ rad/s}$

$x = 0.03 \sin(12.56 t)$

31

$$x = A \sin(2\pi/T \times T/8) = A \sin 45^\circ = A / \sqrt{2}$$

$$PE = \frac{1}{2} k A^2 / 2 = E / 2 \rightarrow KE = E / 2$$

$$KE : PE = 1 : 1$$

a) 1 : 1

II $\omega^2 = a_{\max} / A = 10 / 0.025 = 400 \rightarrow \omega = 20 \text{ rad/s}$

$$f = \omega / 2\pi = 20 / 6.28 = 3.18 \text{ Hz}$$

a) 3.18 Hz

32

A larger amplitude gives a larger restoring force and a larger acceleration, so the longer path is covered faster in the same time : $T = 2\pi \sqrt{m/k}$ contains no A .

T depends only on m and k

II There the potential energy is zero, so all the energy is kinetic.

$PE = 0 \rightarrow KE$ is maximum

III $T = 2\pi \sqrt{m/k}$ does not contain g : the mass and the spring do not change.

T does not depend on g

33

$$\omega = \sqrt{(100 / 1)} = 10 \text{ rad/s}$$

$$v = \omega \sqrt{(A^2 - x^2)} = 10 \times \sqrt{(0.04 - 0.0144)} = 10 \times 0.16 = 1.6 \text{ m/s}$$

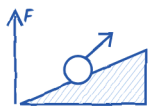
b) 1.6 m/s

II $KE = \frac{1}{2} m v^2 = \frac{1}{2} \times 1 \times (1.6)^2 = 1.28 \text{ J}$

$$PE = \frac{1}{2} k x^2 = \frac{1}{2} \times 100 \times (0.12)^2 = 0.72 \text{ J}$$

$$KE + PE = 2 \text{ J} = \frac{1}{2} k A^2 = \frac{1}{2} \times 100 \times (0.2)^2 \checkmark$$

$KE = 1.28 \text{ J}$, $PE = 0.72 \text{ J}$



34

$$\frac{1}{2} m v_{\max}^2 = E \rightarrow v_{\max} = \sqrt{(2 \times 0.9 / 0.2)} = 3 \text{ m/s}$$

$$v_{\max} = 3 \text{ m/s}$$

I $E = \frac{1}{2} k A^2 \rightarrow k = 2E / A^2 = 1.8 / 0.0225 = 80 \text{ N/m}$

$$k = 80 \text{ N/m}$$

III $T = 2\pi \sqrt{(m / k)} = 6.28 \times \sqrt{(0.2 / 80)} = 6.28 \times 0.05 = 0.314 \text{ s}$

$$a) 0.314 \text{ s}$$

35

$$f \propto 1 / \sqrt{m} \rightarrow f' = f / \sqrt{4} = f / 2$$

$$c) f / 2$$

I $k = F / x \rightarrow \text{newton / metre} = \text{N/m}$

$$b) \text{N/m}$$

III After one complete oscillation the body returns to its starting point, so the displacement is zero (the distance travelled is $4A$).

$$d) \text{zero}$$

36

$$3 = 6 \sin(\pi t) \rightarrow \sin(\pi t) = 0.5 \rightarrow \pi t = \pi / 6$$

$$t = 1/6 \text{ s} \approx 0.167 \text{ s}$$

$$b) 1/6 \text{ s}$$

I $v = \omega \sqrt{(A^2 - x^2)} = \pi \sqrt{(36 - 9)} = 3.14 \times 5.2$

$$v \approx 16.3 \text{ cm/s}$$

$$v \approx 16.3 \text{ cm/s}$$

37

$$v_{\max} = \omega A \rightarrow \omega = 0.6 / 0.03 = 20 \text{ rad/s}$$

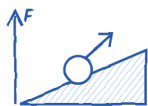
$$d) 20 \text{ rad/s}$$

I $T = 2\pi / \omega = 6.28 / 20 = 0.314 \text{ s}$

$$T = 0.314 \text{ s}$$

III $a = -\omega^2 x = -400 \times 0.015 = -6 \text{ m/s}^2$

$$a = -6 \text{ m/s}^2$$



$T = \text{time} / \text{number of oscillations} = 8 / 20 = 0.4 \text{ s}$

d) 0.4 s

I $k = 4\pi^2 m / T^2 = 39.44 \times 0.4 / 0.16$
 $k \approx 98.6 \text{ N/m}$

$k \approx 98.6 \text{ N/m}$

III $\omega = 2\pi / T = 15.7 \text{ rad/s}$
 $v_{\max} = \omega A = 15.7 \times 0.05 \approx 0.785 \text{ m/s}$

$v_{\max} \approx 0.785 \text{ m/s}$



$|a| = \omega^2 x \rightarrow \omega^2 = 8 / 0.02 = 400$
 $\omega = 20 \text{ rad/s}$

d) 20 rad/s

I $x = |a| / \omega^2 = 12 / 400 = 0.03 \text{ m}$

$x = 3 \text{ cm}$

III $T = 2\pi / \omega = 6.28 / 20 = 0.314 \text{ s}$

$T = 0.314 \text{ s}$



$\omega_1 = \sqrt{(100 / 1)} = 10 \text{ rad/s}$, $\omega_2 = \sqrt{(400 / 1)} = 20 \text{ rad/s}$
 $f_1 / f_2 = \omega_1 / \omega_2 = 1/2$

d) $1/2$

I $E = \frac{1}{2} k A^2$, with the same $A \rightarrow E \propto k$
 $E_1 / E_2 = 100 / 400 = 1/4$
 $(E_1 = 0.125 \text{ J}, E_2 = 0.5 \text{ J})$

$E_1 / E_2 = 1/4$

III $v_{\max} = \omega A \rightarrow v_1 / v_2 = 10 / 20 = 1/2$
 $(v_1 = 0.5 \text{ m/s}, v_2 = 1 \text{ m/s})$

$1/2$